

DryRun

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Breathing Intelligence: Morphological Computation in Inflatable Robot Systems

Sheen Cao

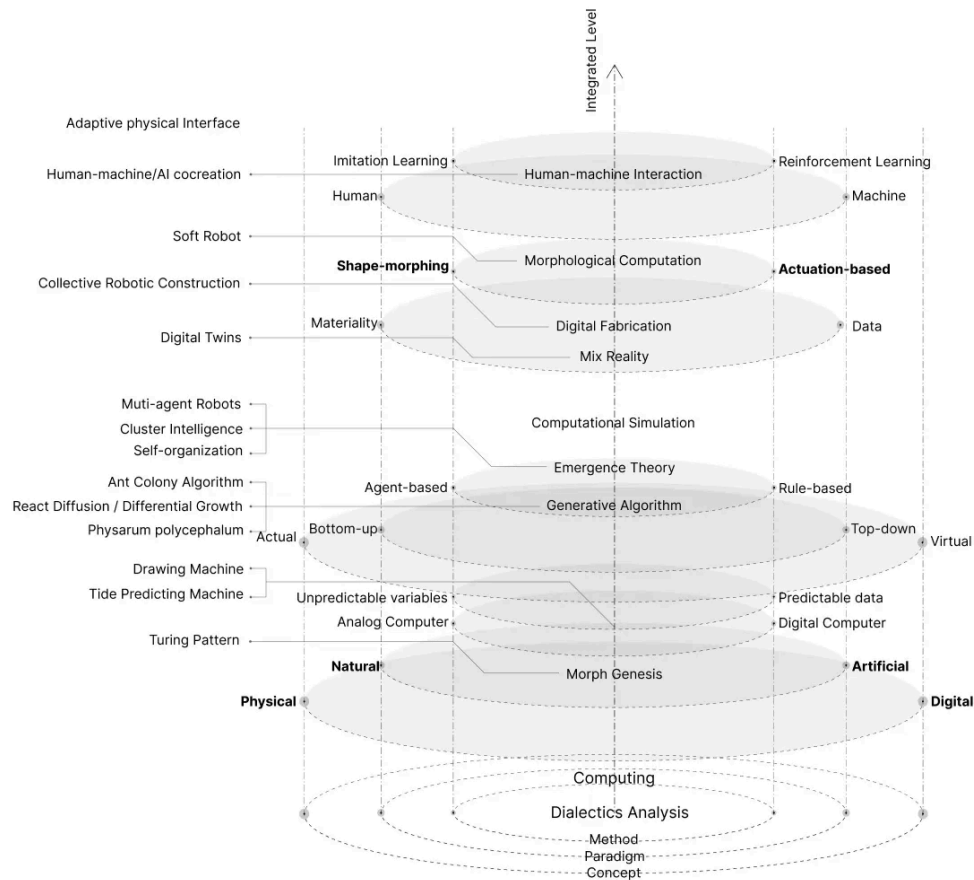
Advisors:

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4/17/2026 Dry Run

Context and background

▼ Dialectics conceptual mapping



- finding it a bit hard to navigate — moving fast and needs a bit more structure in how you bring us into the work.

▼ Key concept pairs

Natural vs. Artificial

Analog vs. Digital

Predictable vs. Unpredictable

Bottom up vs. Top down

Soft vs. Rigid

“Conventionally, engineers have employed rigid materials to fabricate precise, predictable robotic systems, which are easily modeled as rigid members connected at discrete joints. Natural systems, however, often match or exceed the performance of robotic systems with deformable bodies.”

— Danila Rus, Design, fabrication and control of soft robots, 2015

What I am working on?

An inflatable modular robot system that can locomote by coordinating the deforming states.



Related Works

▼ 2026, Yu et al.

Agile Legged Locomotion in Reconfigurable Modular Robots

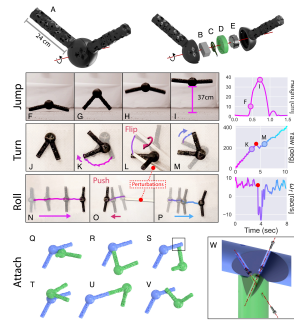


▼ details

▼ 2022, Chin et al.

Auxbot (Flipper-Style Locomotion Through Strong Expanding Modular Robots)

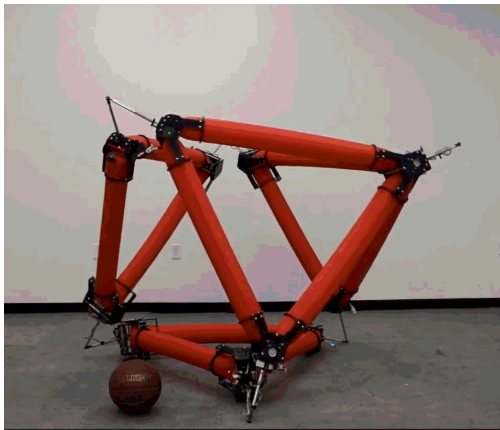




- RL for control policy.
- Bottom-up paradigm

▼ 2020, Usevitch et al.

An untethered isoperimetric soft robot

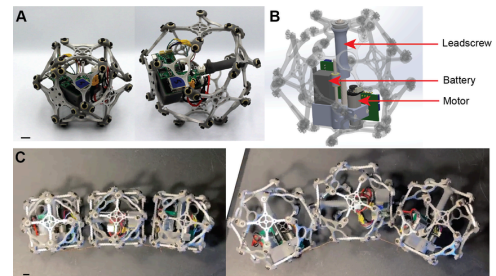


- Geometry driven locomotion

▼ 2010, Wait et al.

Self locomotion of a spherical rolling robot using a novel deformable pneumatic method

▼ details



- con - Complex structural design only for expanding in a limited range.
- pro - Reconfigurability and weight holding capability

▼ 2018, Nozaki et al.

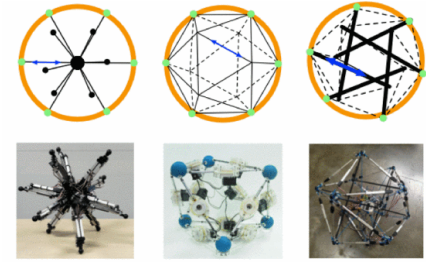
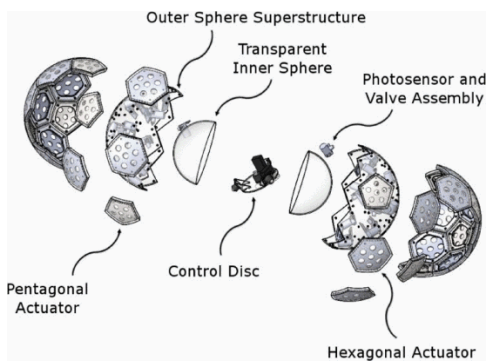
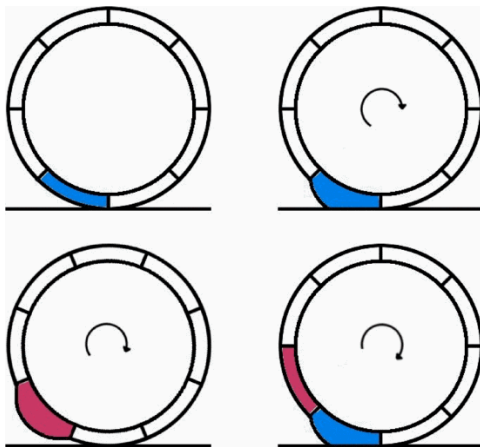
Continuous Shape Changing Locomotion of 32-legged Spherical Robot



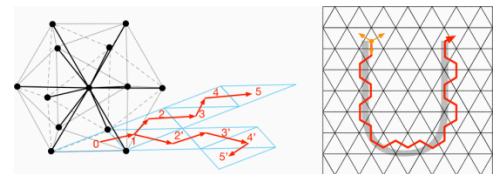
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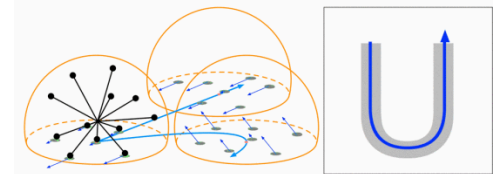
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Radial Skelton [14] Edge Skelton [3] Tensegrity [5]



(a) Discrete Rolling Locomotion



(b) Free Form Locomotion

Literature Positioning

Work	What they do	What's missing
Wait et al. 2010	Inflation → CoM shift → rolling	Single body, no modularity
Usevitch et al. 2020	Soft inflatable + octahedron structure	Not modular, no topology comparison
Chin et al. 2022	Modular volume-changing robot	Rigid, not soft, Limits of expansion, Low mobility
Nozaki et al. 2018	Polyhedral structure for locomotion	Rigid, maximizes DOF (32 is a lot)
Yu et al. 2026	Module count as design variable + RL	Rigid, no geometric constraint

Research Question

take the word cost out -

modularity?

Fundamental Question:

*How can the simple inflation/deflation of balloons, assembled into a particular **geometric configuration**, generate effective locomotion when their **changes are coordinated**?*

Sub questions:

*Configuration: Is there a certain **geometrical configuration** that can balance the mobility and the cost of actuators.*

*Coordination: What is the effective **control policy** to coordinate the states changing of modules to enable the effective locomotion? (adaptive ? how cp for pneu based for locomotion and material ? what needs to be solve?*



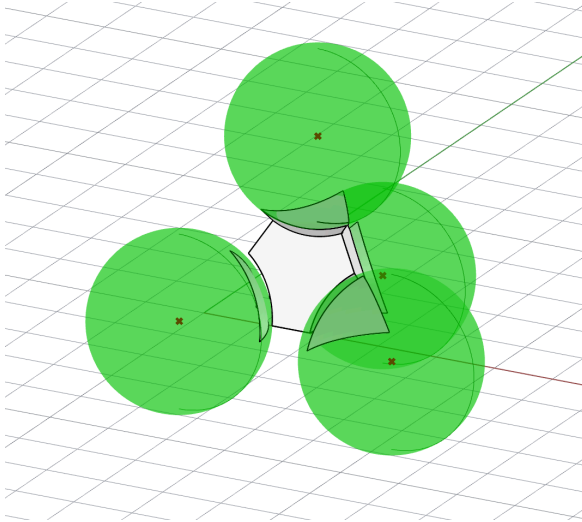
Questions — might need a little refinement — specific to a goal and not really in the context of what you're responding to.

Method and Experiment

Initial prototype

Why Inflatable?

Simple with control as the actuator.



[attachment:0a6ab970-12b9-4ff7-90be-5cc1db010e99:Recording_2026-01-22_142025.mp4](#)

Why central-symmetric?

balance-free, the structure can balance itself

What's the geometry of the core connector?

Compared to biped or quadruped robots, the central symmetric configuration has less trouble with balancing itself, while it's more difficult to gain mobility (starfish can be an example).

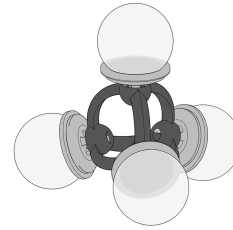
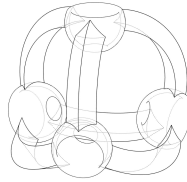
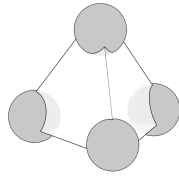
In front of this typical trade-off, there might be a certain configuration that balances the mobility and actuator cost.

Core Piece Structure:

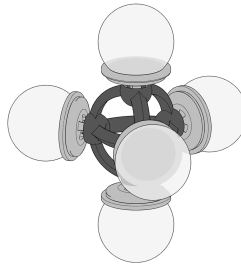
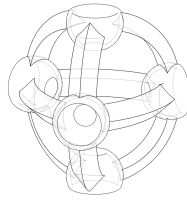
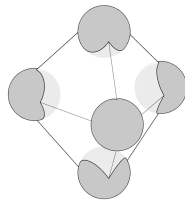
All the vertices are cospherical. — The geometry has a circumscribed sphere.
uniform radius

Configuration Types:

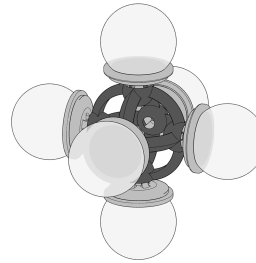
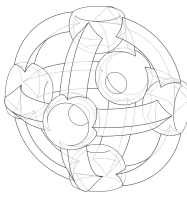
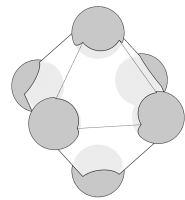
Tetrahedron - 4 units



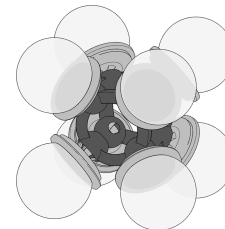
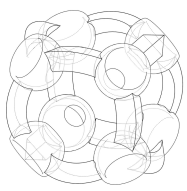
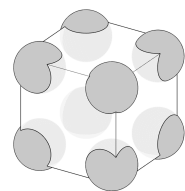
Bipyramid - 5 units



Octahedron - 6 units

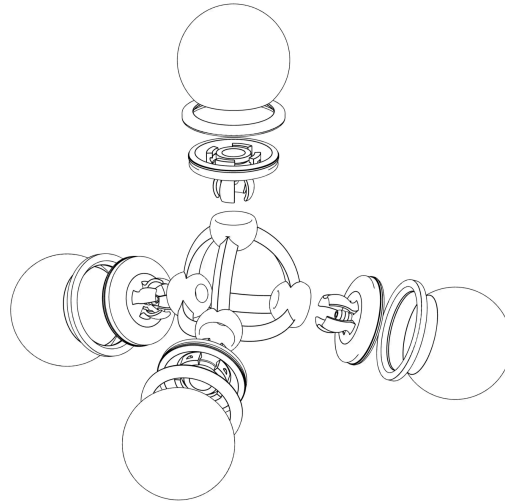
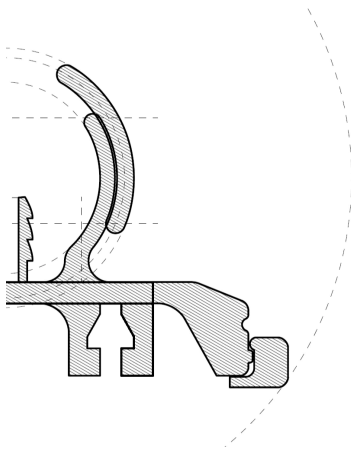


Cube - 8 units



What is the minimum number of inflatable units that can enable effective locomotion?

Holding Plate Design:



For each inflatable module (balloon), I defined 3 different states:

D - deflate

I - Inflate

H - Hold

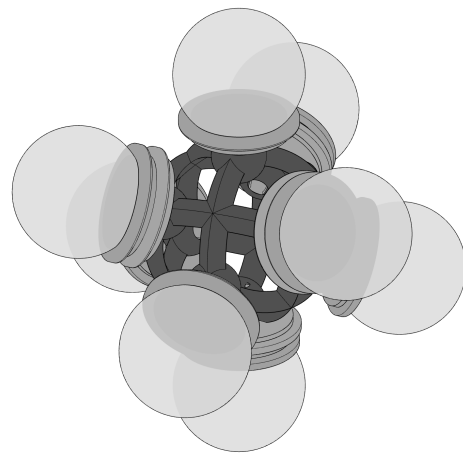
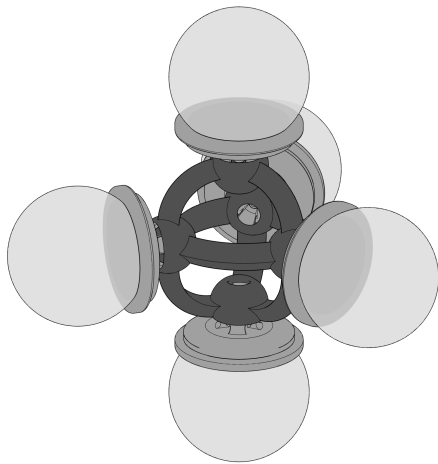
Tetra - Lack of mobility

For the tetrahedron structure, even though each inflatable module can inflate over 1.5 times scale of the core, the whole structure still couldn't reach to an unbalanced state to flip or gait.

[attachment:98dd1e8b-b04b-4b51-967b-4044ef478eb5:IMG_8950.mov](#)

More potential structures:

8-



The gait policy for cube configuration is relatively obvious for 8-module structure.

The inflation on the one branch is easier to cause unbalance.

Config	#Modules	Tip-over capable?	Gait obvious?	Material sensitivity	Selected?
Tetra	4	No	—	—	No (lower bound)
Bipyramid	5	?	?	?	?
Octa	6	Yes	No	High	Yes
Cube	8	Yes	Yes	Low	No (upper bound)



Certain target - simulation and physical - cross target -

[attachment:ad1c869c-14cd-4079-8d62-5728473d6f4f:IMG_9294.mov](#)

[attachment:02decce6-ca32-4b26-8c36-6b77c2167034:IMG_9296.mov](#)



Diagram and formulas about the actuation strategy and geometry.

- Must be shown.
 - minimal threshold.
 - primary: move towards a target and get there - this as a threshold
 - secondary: modulating behavior, speed/gait, etc

1. Library of the geometrical configuration?